

MTH204 (Section A)
Graded Worksheet–1

Name:

Roll No.:

1. **Draw the direction field for the ODE:**

$$y' = 2y - y^2$$

in an integer grid centered at $(0, 0)$ of size 2.

2. **Apply Euler's method to the ODE**

$$y' = y$$

with $h = 0.1$ and $y(0) = 1$, and find three iterations.

3. Let $C(t)$ be the concentration of a drug in the bloodstream. As the body eliminates the drug, $C(t)$ decreases at a rate that is proportional to the amount of the drug that is present at the time. If C_0 is the concentration at time $t = 0$, find the concentration at time t .

4. Solve:

$$(xy^2 - x^2) dx + (3x^2y^2 + x^2y - 2x^3 + y^2) dy = 0.$$

5. Solve:

$$(x^2y^3 + xy) \frac{dy}{dx} = 1.$$

Solution:

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